FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

ENEE4403

POWER SYSTEMS

project

Prepared by:

Karim taha 1211155

Supervised by:

Dr.Jaser Sa'Ed

January , 15, 2025

Abstract

This project aims to take knowledge about power simulator this program is used to implement high voltage systems it shows all the necessary data (transmission impedance, bus voltage, power losses, and the Y bus matrix) it helps to analyze the circuit and see the fault currents and voltage impacts and detect the condition of the system when the load is changed

Contents

Table of Figures

List of Table

Production

System Connection

the bus setting of the slack bus where the bus can be added by inserting the bus from the draw and fill the required information from name, nominal voltage

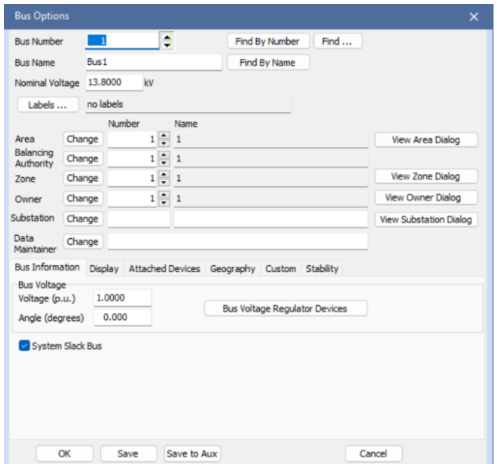


Figure 1:bus option

The generator can be implemented by using the draw option and the required data must be added

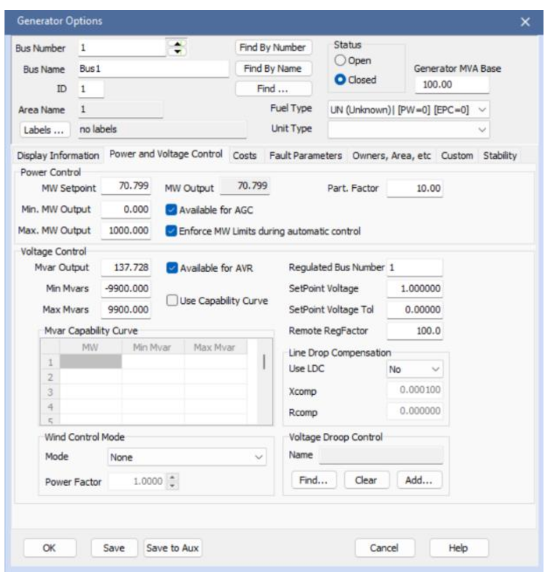


Figure 2:generator settings

The transformer setting

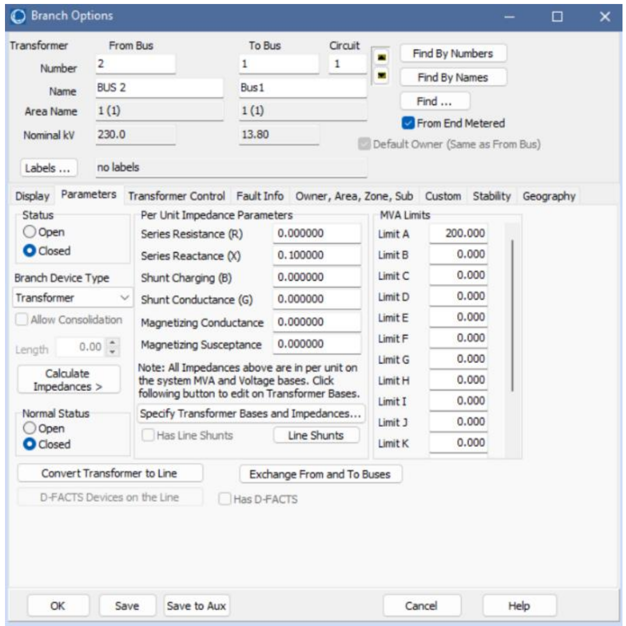


Figure 3:transformer setting

Question1: calculate per unit impedances for all transmission lines

calculating per-unit impedances

the base impedance was calculated to find the value of pre-unit impedance manually and it can be done without calculating using the per unit calculator in the power word simulator when the line be added

$$Z_{PU} = \frac{Z_{actual}}{Z_{base}}\ ,\ Y_{PU} = \frac{Y_{actual}}{Y_{base}}$$

$$Z_{base} = \frac{V_{rated}^2}{S_{base}} = \frac{230^2}{100} = 529$$

$$Y_{base} = \frac{1}{Z_{base}} \quad 1.890*10\text{^}{-3}$$

the calculator impedance of the power word simulator was used to calculate the parameter of the transmission line in the up

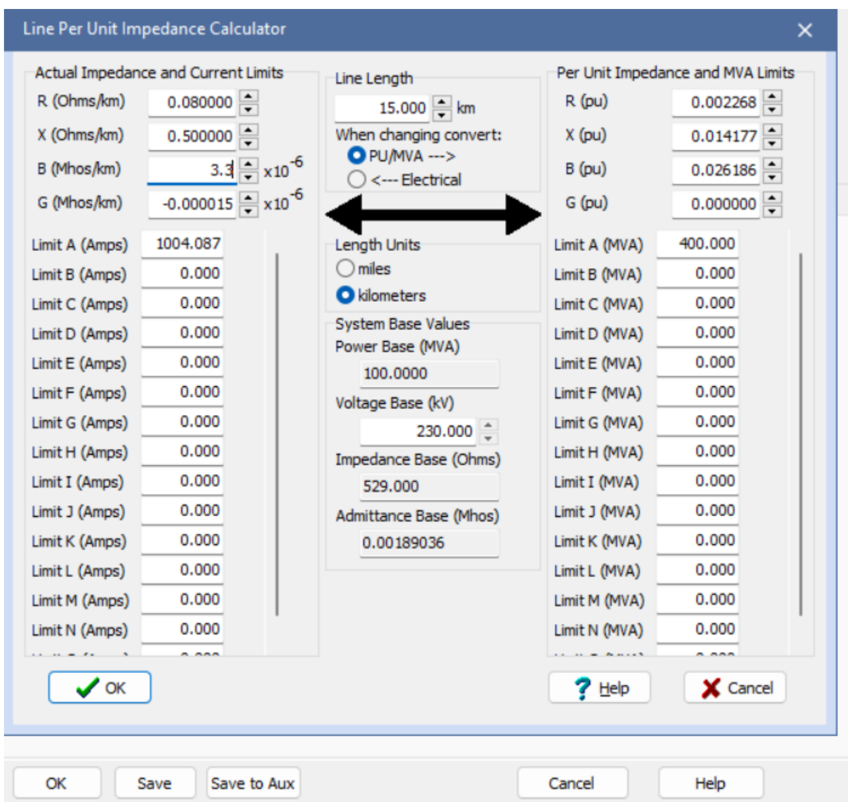


Figure 4: power word parameter calculator

lines parameter

The per-unit impedance of the L2 was calculated manually as shown below:

$$Z_{pu} = \frac{Z_{\text{Actual}}}{Z_{\text{base}}} = \left(\frac{0.08+j0.5}{529}\right)*28 = 0.04233+j0.026459$$

The values of the line parameter of each line are shown in the figure below:

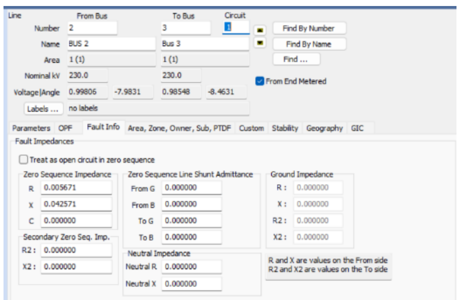


Figure 5:L1 parameter

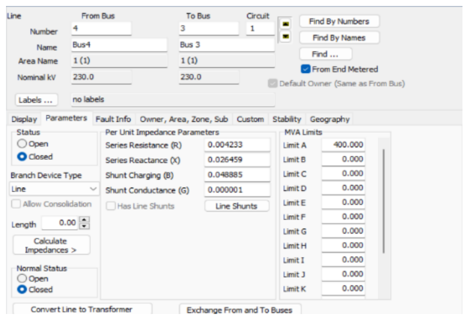


Figure 7:L2 parameter

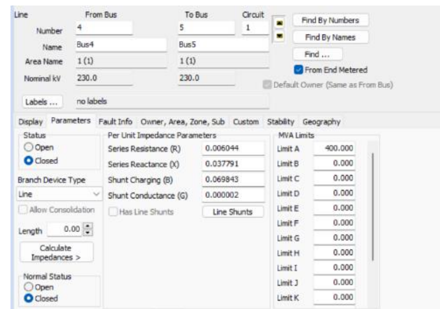


Figure 6:L3 parameter

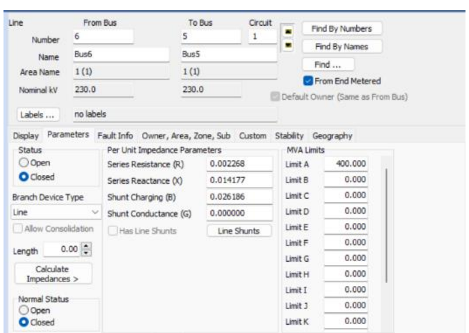


Figure 9:L4 parameter

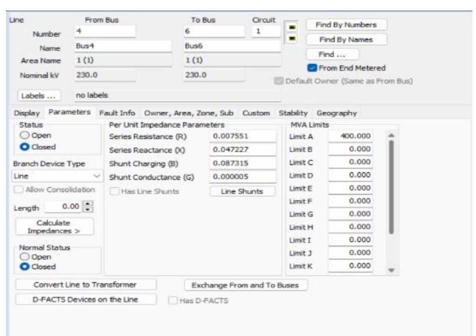


Figure 8:L5 parameter

The parameters of each line are shown in the table below this table can be found in case >>>> branch input the value of each parameter depends on length of each line

Table 1:Transmtion line parameters Table

	To Number	To Name	Circuit	Status	Branch Device Type	Xfrmr	R	X	B	Lim MVA A
1	3	Bus 3	1	Closed	Line	NO	0.00423	0.02646	0.04888	400.0
2	3	Bus 3	1	Closed	Line	NO	0.00227	0.01418	0.02619	400.0
3	1	Bus1	1	Closed	Transformer	YES	0.00000	0.10000	0.00000	200.0
4	5	Bus5	1	Closed	Line	NO	0.00227	0.01418	0.02619	400.0
5	5	Bus5	1	Closed	Line	NO	0.00604	0.03779	0.06984	400.0
6	6	Bus6	1	Closed	Line	NO	0.00755	0.04723	0.08732	400.0
7	7	Bus7	1	Closed	Transformer	YES	0.00000	0.10000	0.00000	200.0

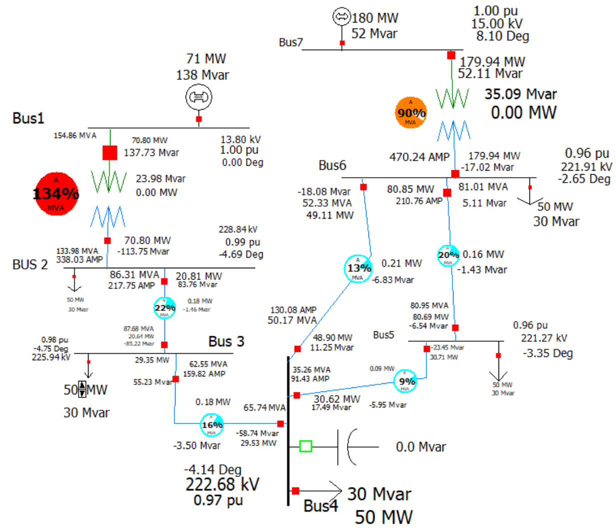


Figure 10:system as the required data

As noted transformer one is fully loaded since the total loaded power is 250 MW and 150 MVar and the generation power is equal to 250.8 MW and 189Mvar so this surplus will affect the complex power that is on the slack bus is higher than the rated value on the transformer so the transformer will be full loaded so the value of the complex power of the transformer was changed to 200MVar

Question 2-part a

One-line diagram with data

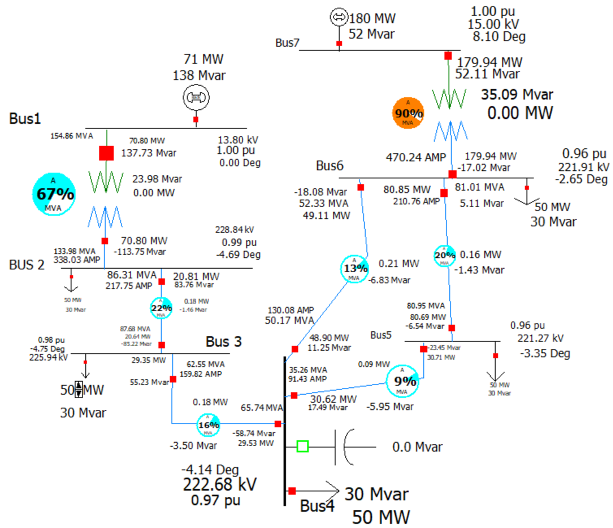


Figure 11:system structure

running the system

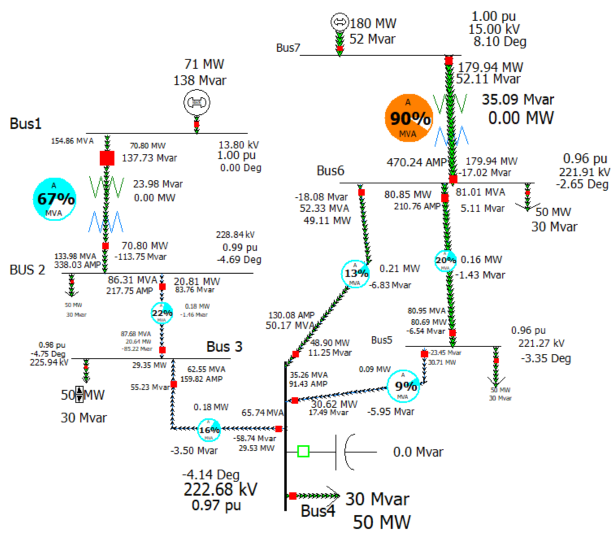


Figure 12:running the

Question 2-part b

Total load, total generation power, and total losses this data can be seen from the case summary:

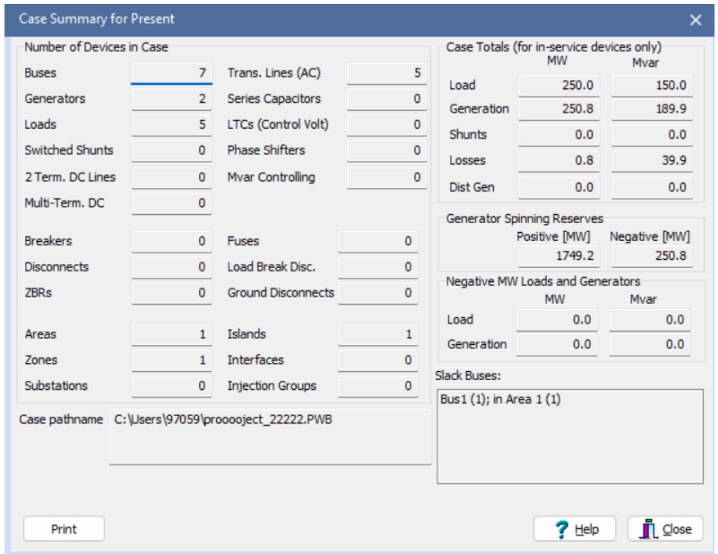


Figure 13:Total load, power and losses

Question 2-part c

The y bus matrix can be found from the solution detail (Y bus matrix)

the table of Y bus matrix can be found from >>>solution detail >>> Y bus matrix

Table 2:Y Bus Matrix

	Number	Name	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5	Bus 6	Bus 7
1	1	Bus1	0.00 - j10.00	-0.00 + j8.70					
2	2	BUS 2	-0.00 + j8.70	11.00 - j76.32	-11.00 + j68.78				
3	3	Bus 3		-11.00 + j68.78	16.90 - j105.59	-5.90 + j36.85			
4	4	Bus4			-5.90 + j36.85	13.32 - j83.20	-4.13 + j25.80	-3.30 + j20.65	
5	5	Bus5				-4.13 + j25.80	15.13 - j94.53	-11.00 + j68.78	
6	6	Bus6				-3.30 + j20.65	-11.00 + j68.78	14.30 - j99.37	-0.00 + j10.00
7	7	Bus7						-0.00 + j10.00	0.00 - j10.00

Question 2 -part d

Table 3:Busess data

	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Load MW	Load Mvar	Gen MW	Gen Mvar	Switched Shunts Mvar	Act G Shunt MW	Act B Shunt Mvar	Area Num	Zone Num
1	1	Bus1	1	13.80	1.00000	13.800	0.00			70.82	137.77		0.00	0.00	1	1
2	2	BUS 2	1	230.00	0.99490	228.827	-4.70	50.00	30.00				0.00	0.00	1	1
3	3	Bus 3	1	230.00	0.98230	225.930	-4.76	50.00	30.00				0.00	0.00	1	1
4	4	Bus4	1	230.00	0.96811	222.666	-4.14	50.00	30.00				0.00	0.00	1	1
5	5	Bus5	1	230.00	0.96201	221.262	-3.35	50.00	30.00				0.00	0.00	1	1
6	6	Bus6	1	230.00	0.96477	221.897	-2.65	50.00	30.00				0.00	0.00	1	1
7	7	Bus7	1	15.00	1.00000	15.000	8.10			180.00	52.17		0.00	0.00	1	1

Question 2-part e

Table 4:Branches state

Branches State | Buses | DC Lines | Branches Input | Ybus

Records Geo Set Columns f(x) Options

Filter Advanced Branch Find... Remove Quick Filter

	From Number	From Name	To Number	To Name	Circuit	Status	Branch Device Type	Xfrmr	MW From	Mvar From	MVA From	Lim MVA	% of MVA Limit (Max)	MW Loss	Mvar Loss
1	2	BUS 2	1	Bus1	1	Closed	Transforme	YES	-70.8	-113.8	134.0	200.0	77.5	0.00	24.00
2	2	BUS 2	3	Bus 3	1	Closed	Line	NO	20.8	83.8	86.3	400.0	21.9	0.18	-1.46
3	4	Bus4	3	Bus 3	1	Closed	Line	NO	29.5	-58.7	65.7	400.0	16.4	0.18	-3.50
4	4	Bus4	5	Bus5	1	Closed	Line	NO	-30.6	17.5	35.3	400.0	9.7	0.09	-5.95
5	4	Bus4	6	Bus6	1	Closed	Line	NO	-48.9	11.2	50.2	400.0	13.1	0.21	-6.83
6	6	Bus6	5	Bus5	1	Closed	Line	NO	80.9	5.1	81.0	400.0	20.3	0.16	-1.43
7	6	Bus6	7	Bus7	1	Closed	Transforme	YES	-180.0	-17.0	180.8	200.0	93.7	0.00	35.12

This table shows the apparent power (real power +j reactive power) that has been dissipated in the transmission line and passes through the transformers and the limit of MVA the reactive power losses much higher than the real power since the long distance of the transmission line and the high current that pass through the transmission line.

The direction of the power value is determined by the power angle, transitioning from a higher power angle to a lower one. The current value in a transmission line varies due to differences in line length, with Zpu being influenced by this length. Moreover, an increase in the transmission line's length leads to greater losses, as Z rises in proportion to the length, resulting in a reduction of current within the line. The power in the line is influenced by the power angle, with power flow remaining stable for $\delta<90^\circ$ and becoming unstable for $\delta>90$. This simulation offers the voltage and angle for each bus, in addition to the quantity of power (both real power in MW and reactive power in MVar) utilized at every bus. It likewise illustrates how much power is lost within the system, both real and reactive, as well as the current that flows through the transmission lines and transformers.

Question 3 Fully loaded Transformer

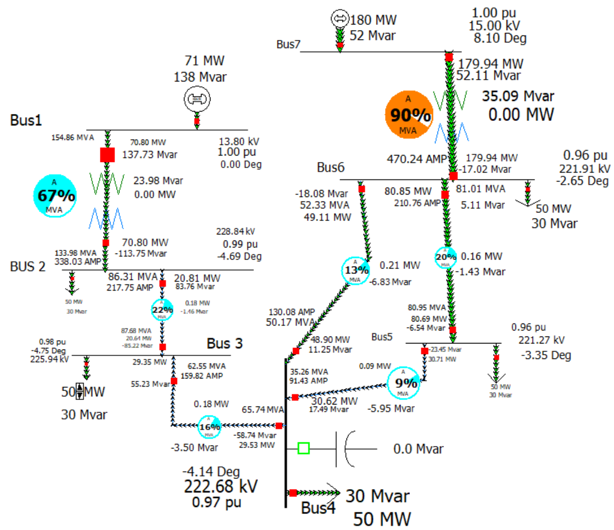


Figure 14: System before increasing the load

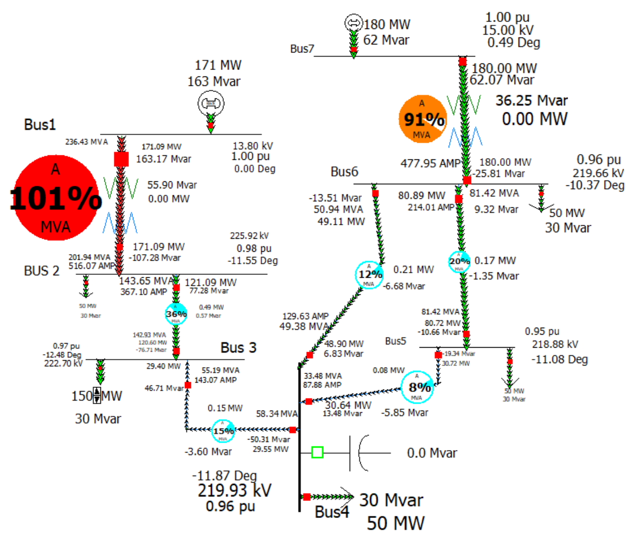


Figure 15:svstem when the transformer T1 overloaded

The value that makes the T1 fully loaded is 150MW

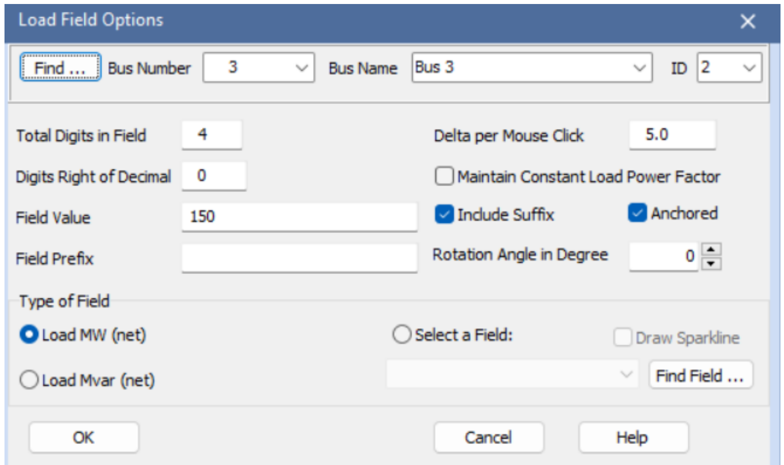


Figure 16: the value of the load on Bus 3 when T1 is fully loaded

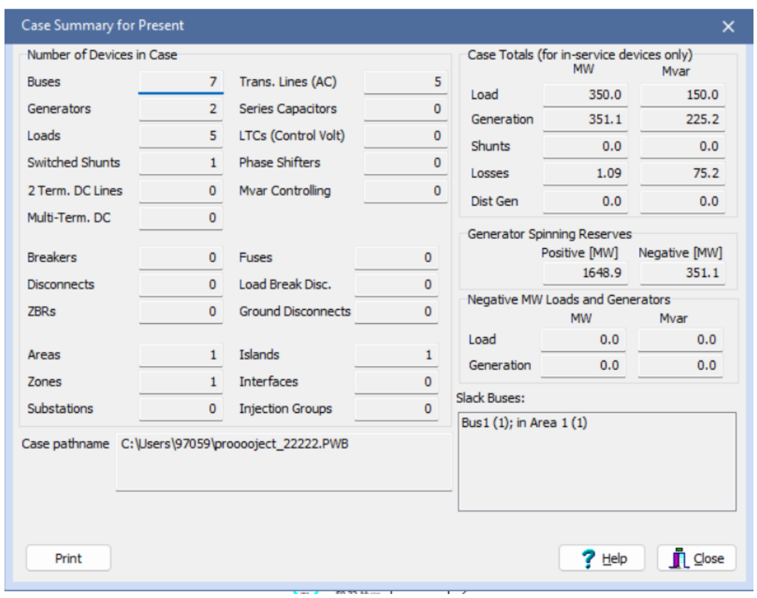


Figure 17:power generator and losses at full loaded

The losses of the transmission line increased since the current in the transmission increased which directly affects the losses ($P=I^2R$) when the load at Bus 3 rises to 150 MW, the voltage at Bus 3 along with other buses drops due to the increased current. Transmission lines and transformers near Bus 3 can handle more load, increasing power losses and the risk of overload. Generators must supply additional power, which may strain the system without a reactive power supply, voltage stability may be compromised. as losses increase, overall system efficiency decreases

Question 4 shunt capacitor

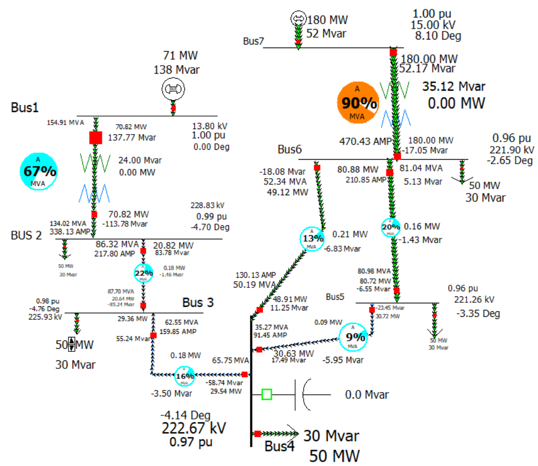


Figure 18:before the capacitor

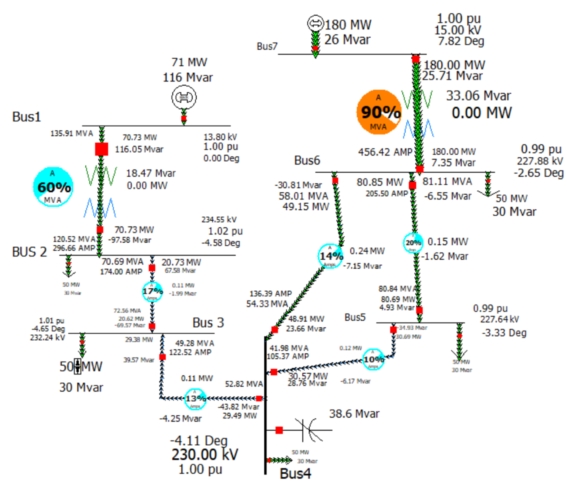


Figure 19: after adding the capacitor

$$C = \frac{Qc}{V^2 w} = \frac{38.6*10^6}{(230k)^2\,(2*pi*50)} = 2.323\mu F$$

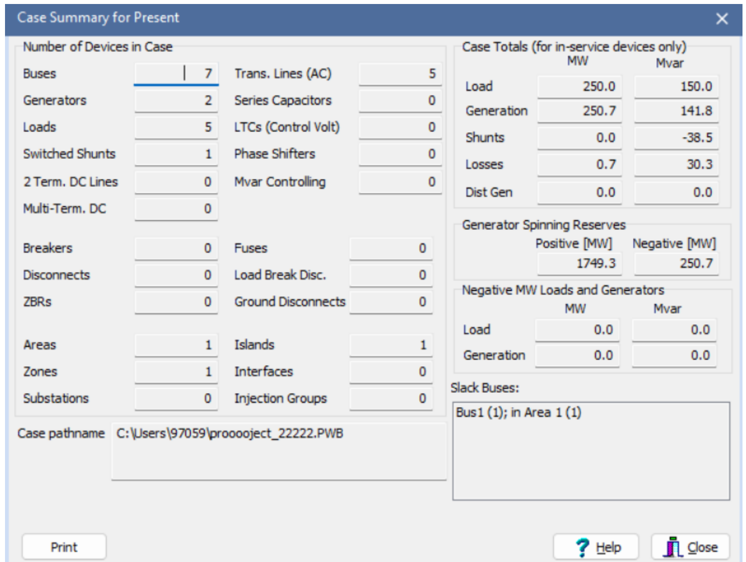
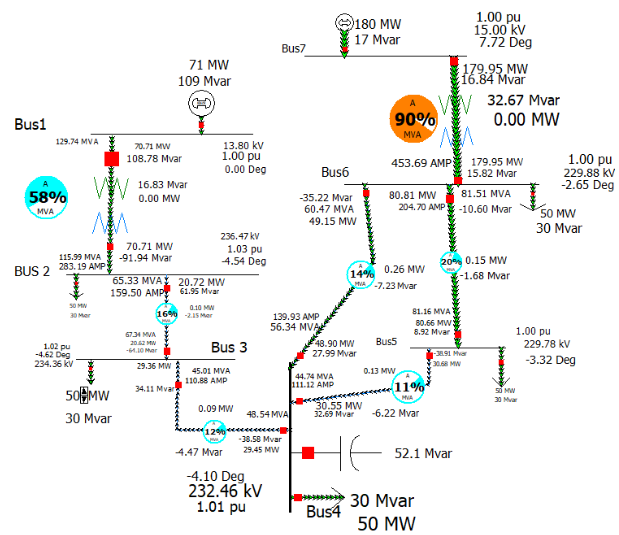


Figure 20:total load generator and loses when shunt =38.6Mvar



\

Figure 21:after increase the value of the capacitor to51Mvar

$$C = \frac{Qc}{V^2 w} = \frac{52.1*10^{\wedge}6}{(230k)^2\ (2*pi*50)} = 3.13\mu\text{F}$$

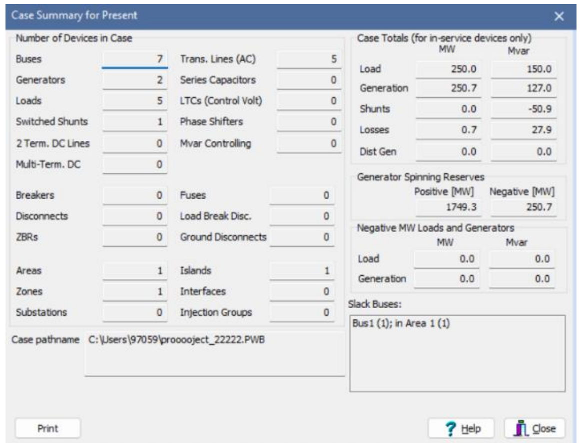


Figure 22:total load generator and loses when the shunt =51MVR

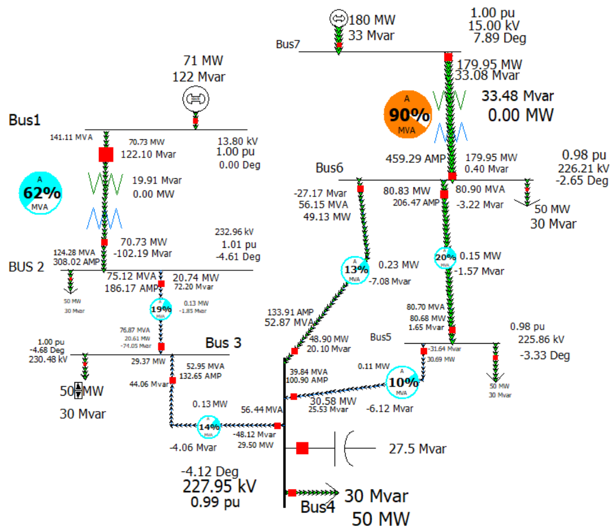


Figure 23:System with shunt =28Mvar

$$C = \frac{Qc}{V^2 w} = \frac{27.5*10^6}{(230k)^2\,(2*pi*50)} = 1.6\mu\text{F}$$

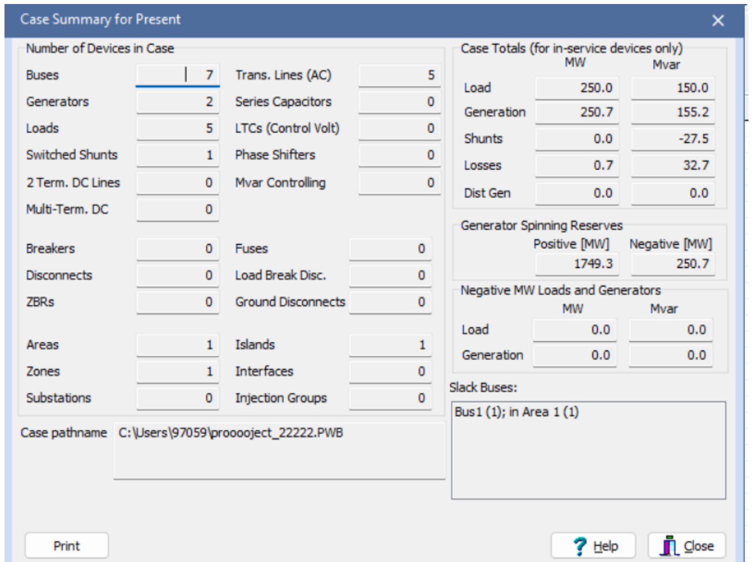


Figure 24:total load when shunt =28Mvar

Adding a capacitor reduces the loaded reactive power which helps to reduce the voltage drop in the transmission line by lowering losses and improving efficiency the capacitor enhanced voltage stability, reduced losses, and improved system performance since the value of Mvar increases in the bus the total reactive power in the network decreases before adding the shunt the value of the reactive power in line two was -58.74Mvar after adding the capacitor it decreases to -43.82Mvar it also help to make the system more stable the required shunt capacitance to make the voltage in bus 4 equal 1 pu is 38Mvar and if the value of the shunt capacitance increases the losses of the system will decrease and the system becomes more stable

Question 5 Fault

to calculate the parameter zo in pu

$$Z_{OPU} = \frac{Z_{O\ Actual}}{Z_{base}} = \frac{0.2+j1.5}{529} * 18 = 0.01058+j0.079338$$

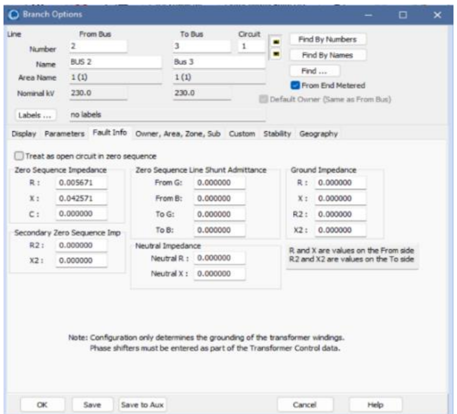


Figure 29:Transmission 1 line fault parameter

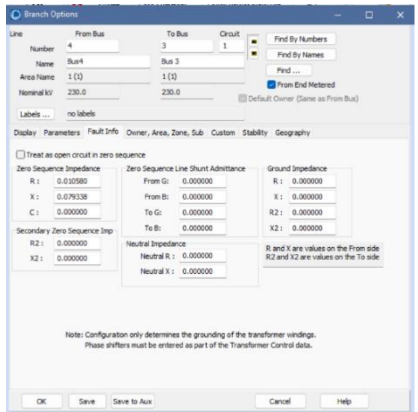


Figure 28:Transmission 2 line fault parameter

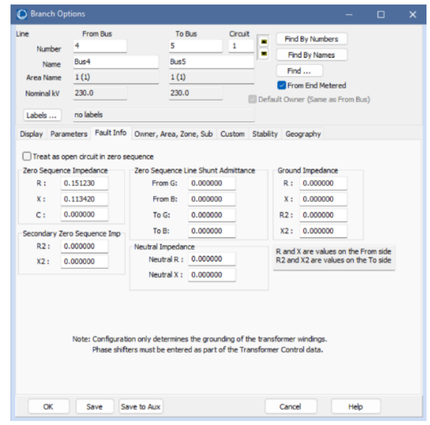


Figure 27::Transmission 3 line fault parameter

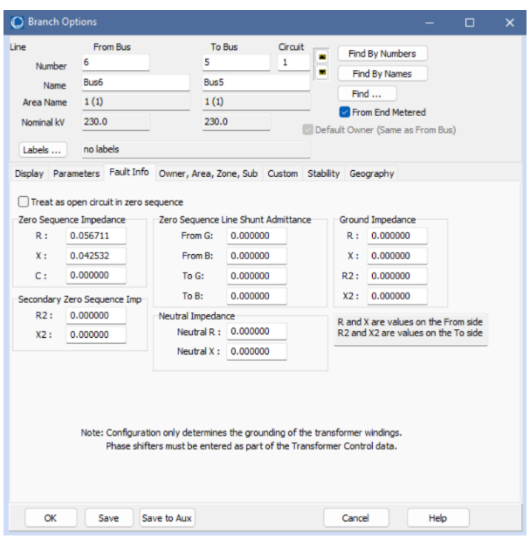


Figure 26:Transmission 4 line fault parameter

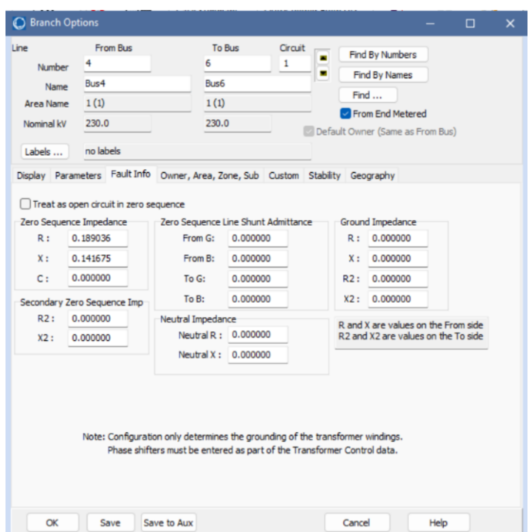


Figure 25:Transmission 5 line fault parameter

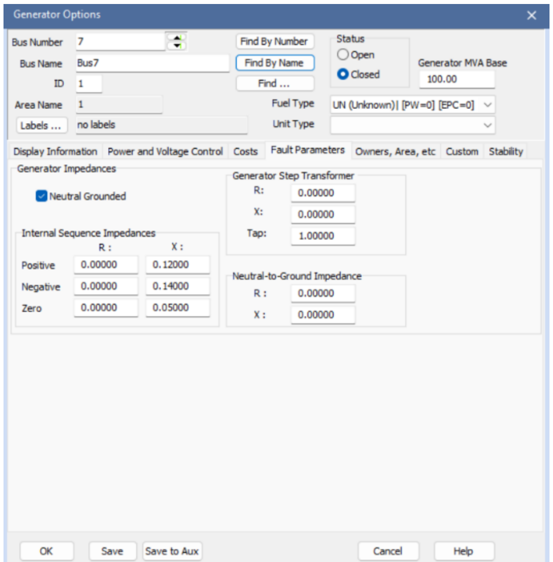


Figure 32:pv generator fault settings

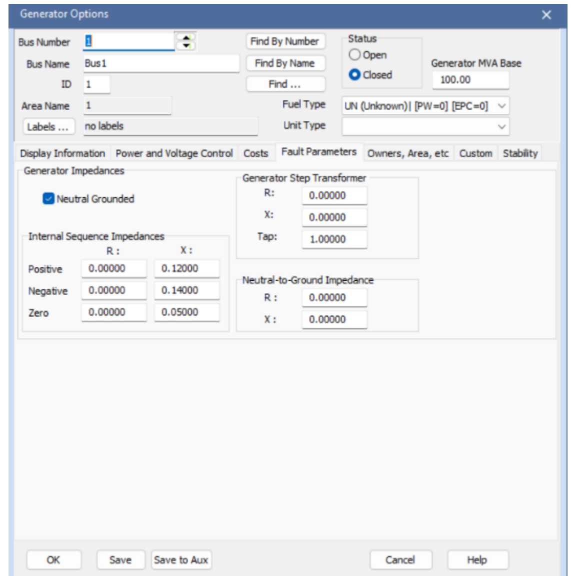
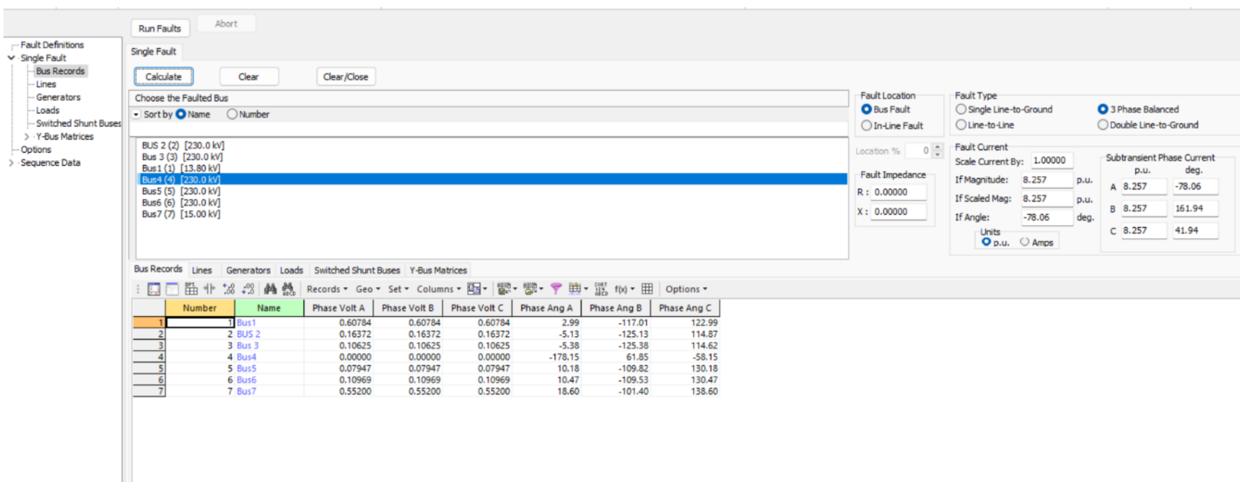


Figure 31:slack generators fault parameter



	Number	Name	Phase Volt A	Phase Volt B	Phase Volt C	Phase Ang A	Phase Ang B	Phase Ang C
1	1	Bus1	0.60784	0.60784	0.60784	2.99	-117.01	122.99
2	2	BUS 2	0.16372	0.16372	0.16372	-5.13	-125.13	114.87
3	3	Bus 3	0.10625	0.10625	0.10625	-5.38	-125.38	114.62
4	4	Bus4	0.00000	0.00000	0.00000	-178.15	61.85	-58.15
5	5	Bus5	0.07947	0.07947	0.07947	10.18	-109.82	130.18
6	6	Bus6	0.10969	0.10969	0.10969	10.47	-109.53	130.47
7	7	Bus7	0.55200	0.55200	0.55200	18.60	-101.40	138.60

Figure 33:bus record while the fault

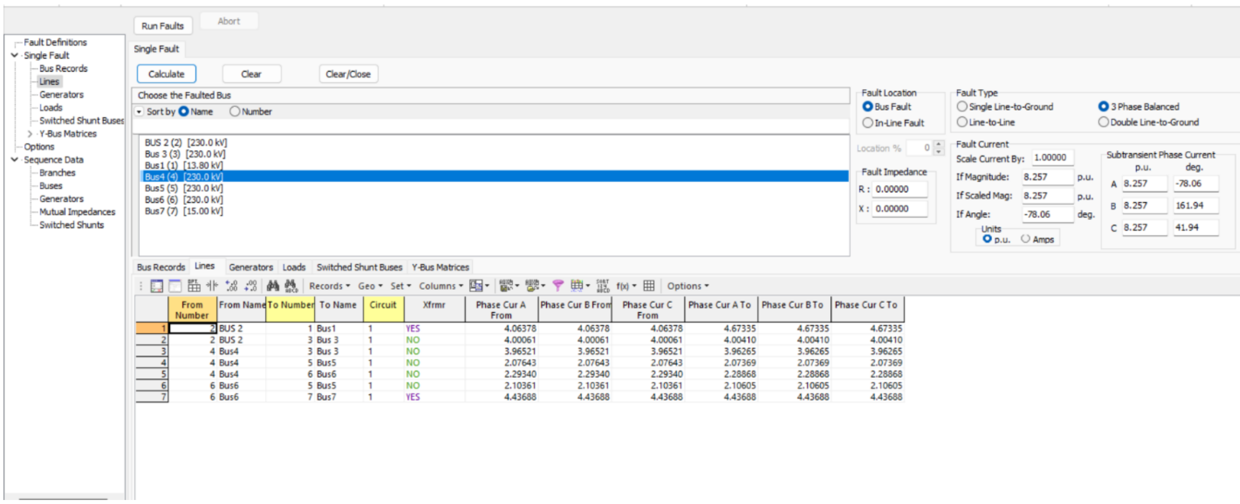


Figure 34: lines recoded

A three-phase balanced fault is applied to multiple buses in the system, resulting in the highest fault current magnitude at bus 4 where the voltage at it becomes 0 and the current 8.257 with angle-78.06 this test helps to take the worst case that can happen to the system and take the necessary transmission line cable and protection that should be used to avoid the current born

Conclusion

The knowledge of the power word simulator was taken, where the system was built, designed, and analyzed using the power word, the Y Bus matrix, power flow, losses, and voltages on buses were determined, and the T1 was set to be fully loaded, and the value of the load that makes it full loaded 150MW and a shunt capacitor was added to bus 4 to see the effect of the shunt capacitance, where it helps the system to be more stable and reduces the losses, a three-phase fault was made on bus 4 and the value of current was determined and the results of the fault on the system was seen and analyzed